The limit of this as $\Delta \mathrm{E} \to 0$ defines a quantity

$$\mu(\mathcal{M}) = \int_{\mathcal{M}} \frac{1}{\|\nabla H\|} \mathrm{d}\sigma.$$

As both volume and energy are invariant under the Hamiltonian flow, $\mu(\mathcal{M})$ must be invariant under the flow, thus

$$\mu(\mathscr{F}_t(\mathcal{M})) = \mu(\mathcal{M}).$$

The invariant measure of the whole level set Σ_{E} is defined as the *microcanonical partition function*

$$Z_{\mathrm{E}} = \mu(\Sigma_{\mathrm{E}}).$$

Example 5.1 Consider the harmonic oscillator $H(q,p) = p^2/2 + \Omega^2 q^2/2$. The energy surface $H = \mathrm{E} > 0$ is an ellipse. The partition function is obtained by integrating in the standard way. Parameterizing the surface by q, we have $p = \sqrt{2\mathrm{E} - \Omega^2 q^2}$, then

$$\begin{aligned}
\mu(\Sigma_{\mathrm{E}}) &= 4 \int_0^{\sqrt{2\mathrm{E}}} \frac{\sqrt{1 + p_q^2}}{\sqrt{H_p^2 + H_q^2}} \mathrm{d}q \\
&= 4 \int_0^{\sqrt{2\mathrm{E}}/\Omega} \frac{\sqrt{1 + \frac{\Omega^4 q^2}{2\mathrm{E} - \Omega^2 q^2}}}{\sqrt{p^2 + \Omega^4 q^2}} \mathrm{d}q \\
&= 4 \int_0^{\sqrt{2\mathrm{E}}/\Omega} \frac{1}{\sqrt{2\mathrm{E} - \Omega^2 q^2}} \mathrm{d}q \\
&= 2\pi/\Omega.
\end{aligned}$$

Example 5.2 Consider a system of N non-interacting particles of mass m moving in periodic boundary conditions within a box of volume V. The Hamiltonian is

$$H = \frac{1}{2m} \sum_{i=1}^{\mathrm{N}} \|\boldsymbol{p}_i\|^2$$

so

$$Z_{\mathrm{E}} = \mathrm{V}^{\mathrm{N}} \int_{\mathrm{S}} \frac{m}{r} \mathrm{d}\sigma$$